Kubernetes Lab Assignment: CampusEats Rollout

Scenario

You are the DevOps intern at CampusEats, a student-run startup that lets hostel residents pre-order from campus canteens, and you must show that its web apps run reliably on Kubernetes.

The developers have containerized two apps: an nginx front end and an Apache (httpd) page for the daily menu. Your team lead has raised five tickets to clear on your own laptop's Docker Desktop cluster.

Ground rules

Work only on your own laptop's cluster, and end every resource name with your roll-number suffix.

- **Tools:** Docker Desktop (you enable Kubernetes in Ticket 1) and `kubectl`.
- **Names:** end every resource name with `-NNN`, the last three digits of your roll number, for example `ce-web-472`.
- **Commands:** none are given, except where a ticket says so. Use your Day 18 notes and the built-in help (for example `kubectl create --help`).
- **Terminals:** a port-forward or a watch (`-w`) keeps running and blocks its terminal, so open another one for the next command. In Windows PowerShell type `curl.exe`, because plain `curl` is a different command there.
- **Safety:** delete only what a ticket tells you to delete.

Ticket 1: Cluster readiness check

Lead: "Before we deploy anything, show me the cluster is healthy and that you are pointed at the right one."

1. Enable Kubernetes in Docker Desktop and wait until it shows as running.
2. Show the client and server versions.
3. List all contexts and make `docker-desktop` the active one.
4. List the nodes, then show the node's wide listing and its full description.

Ticket 2: Smoke test with a bare pod

Lead: “Run the front end as one pod and prove it serves traffic.”

1. Run a pod named `ce-smoke-NNN` from the `nginx` image.
2. Confirm it is Running, then list everything in the namespace.
3. Forward local port 8080 to port 80 of the pod. In a second terminal or your browser, fetch the page and capture the nginx welcome response.
4. Stop the forward, delete the pod, then list everything in the namespace again.

Hint: if port 8080 is busy on your laptop, use another local port.

Ticket 3: Deployments and self-healing

Lead: “One pod is not production. Run both apps as deployments, then break one on purpose and watch what happens.”

1. Create a deployment `ce-web-NNN` from `nginx` with 3 replicas.
2. Create a deployment `ce-menu-NNN` from `httpd` with the default replica count.
3. List the pods with their node and pod IP.
4. For `ce-menu-NNN`, print the YAML and the full description.
5. Start two watchers in separate terminals, one for pods and one for events. In a third terminal, delete one `ce-web-NNN` pod.
6. When the pod count is back to 3, list the pods with their IPs again, then list the events.

Ticket 4: Expose the app

Lead: “Pod IPs change whenever a pod is replaced. Give the front end a stable address, then check who can reach it.”

1. Expose `ce-web-NNN` inside the cluster as a ClusterIP service named `ce-clusterip-NNN` on port 80. Note its cluster IP and its endpoints.
2. Start a helper pod `ce-tester-NNN` from the `curlimages/curl` image that sleeps for an hour. Wait until it is Ready.
3. From inside the helper pod, fetch the page using the service name.
4. From your laptop, fetch the page using the cluster IP, with a 3-second time limit.
5. Expose `ce-web-NNN` as a NodePort service named `ce-nodeport-NNN` on port 80. Note the node port.
6. Fetch the page from your laptop through `localhost` and the node port, then from the helper pod through the NodePort service name.

7. Delete one `ce-web-NNN` pod, then describe the ClusterIP service again.

Ticket 5: Incident, a release that will not start

Lead: "A teammate deployed the payments page and it never came up. Tell me the cause before you touch anything."

1. Run this, replacing NNN with your suffix: `kubectl create deployment ce-pay-NNN --image=nginx:1.99.99`
2. Wait about a minute, then list the pods and note the STATUS.
3. Describe the failing pod and read its Events section. List the namespace events as well.
4. Find the root cause and copy the exact message that proves it.
5. Fix it: remove the broken deployment (`kubectl delete --help` shows how), then create it again from the `nginx` image.
6. Confirm the new pod is Running.

Cleanup

Leave the cluster as you found it: only the built-in `service/kubernetes` should remain.

1. Delete every object in the default namespace (safe only because this is your own laptop cluster).
2. List everything in the namespace again. The built-in service may take a moment to reappear.